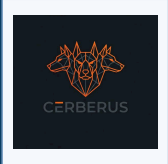


# Credit Spread Leading Indicator



INTERNAL

Use HY/IG OAS and stress flags as a leading indicator for equity drawdowns.

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|                                                     |                                                                       |                                      |
|-----------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------|
| <div>N</div> <div>106</div>                         | <div>EDGE</div> <div>-1.83%</div>                                     | <div>HIT-RATE</div> <div>46.2%</div> |
| <div>CLASSIFICATION</div> <div>internal</div>       | <div>AUTHOR</div> <div>Cerberus Research</div>                        |                                      |
| <div>RUN ID</div> <div>20260807_094711</div>        | <div>GIT COMMIT</div> <div>978cf61e</div>                             |                                      |
| <div>DURATION</div> <div>2.1s</div>                 | <div>DATASETS</div> <div>credit_spreads/credit_spreads_features</div> |                                      |
| <div>PARAMETER · LOOKBACK_YEARS</div> <div>20</div> | <div>PARAMETER · HORIZON</div> <div>63</div>                          |                                      |
| <div>PARAMETER · BENCHMARK</div> <div>SPY</div>     | <div>KPI</div> <div>Value</div>                                       |                                      |
| <div>N</div> <div>106</div>                         | <div>EDGE</div> <div>-1.83%</div>                                     |                                      |
| <div>HIT-RATE</div> <div>46.2%</div>                |                                                                       |                                      |

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| <div>KEY FINDINGS</div> <div><div>· Credit spread vs 63-bar SPY return: <math>r=0.60</math>, <math>p=0.000</math></div></div> |
|-------------------------------------------------------------------------------------------------------------------------------|

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## Data Sources

- `credit_spreads/credit_spreads_features`

## Key Findings

- Credit spread vs 63-bar SPY return:  $r=0.60$ ,  $p=0.000$

## Executive Summary

**NOTE**

Tests whether HY/IG credit spread widening leads equity drawdowns over a multi-month horizon.

## Results & Interpretation

- Across 106 events, the forward return averaged  $-1.83\%$ . Returns were positive 46.2% of the time.
- Across 143 events, the forward return averaged  $+2.56\%$ . Returns were positive 74.1% of the time.
- Across 157 events, the forward return averaged  $+5.62\%$ . Returns were positive 95.5% of the time.
- Across 157 events, the forward return averaged  $+7.26\%$ . Returns were positive 100.0% of the time.
- Across 158 events, the forward return averaged  $+9.42\%$ . Returns were positive 93.0% of the time.

## Objectives

**NOTE**

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

## Methodology

**NOTE**

Analysis is run on the configured datasets with the parameters listed in the Study Configuration block. Results are aggregated into summary statistics, tables, and visual one-pagers for review.

## Charts

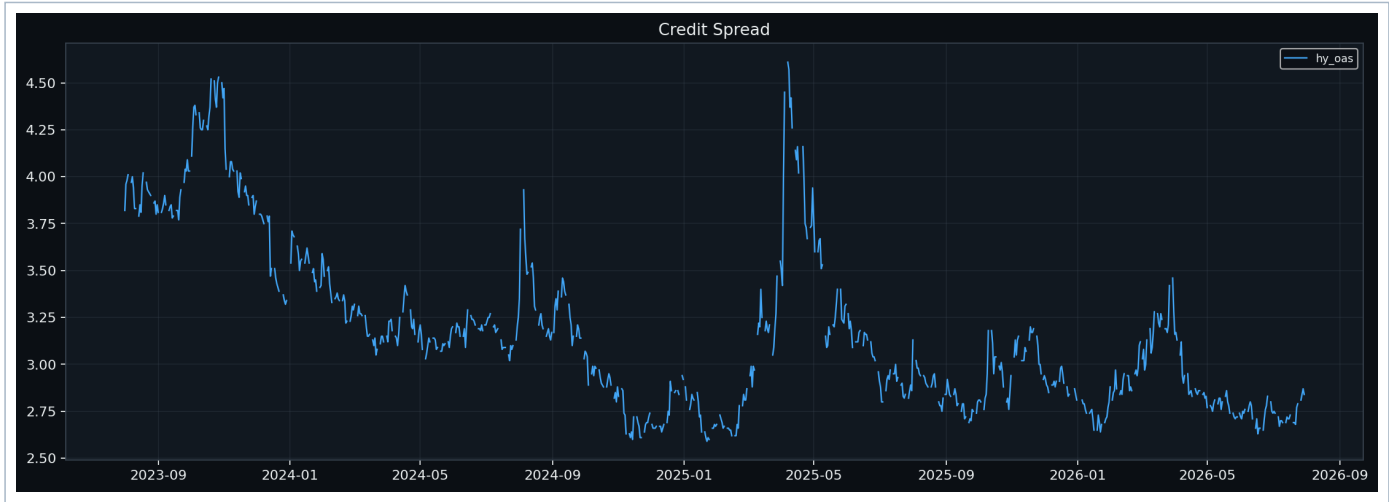


Figure 1. Credit spread over time  
\_Credit spread over time\_

## Statistical Results

| hy_oas |
|--------|
| 2.88   |
| 2.90   |
| 2.94   |
| 2.95   |
| 2.88   |
| 2.80   |
| 2.78   |
| 2.78   |
| 2.75   |
| 2.82   |
| 2.84   |
| 2.84   |
| 2.92   |
| 2.88   |
| 2.84   |
| 2.83   |
| 2.84   |
| 2.87   |
| 2.84   |
| 2.78   |

| hy_oas |
|--------|
| 2.79   |
| 2.75   |
| 2.79   |
| 2.79   |
| 2.71   |
| 2.72   |
| 2.69   |
| 2.71   |
| 2.70   |
| 2.76   |
| 2.75   |
| 2.74   |
| 2.80   |
| 2.81   |
| 2.81   |
| 2.80   |
| 2.76   |
| 2.82   |
| 2.84   |
| 2.95   |
| 3.18   |
| 3.18   |
| 3.11   |
| 2.95   |
| 3.04   |
| 3.04   |
| 2.99   |
| 2.97   |
| 3.01   |
| 2.96   |
| 2.88   |
| 2.80   |

| hy_oas |
|--------|
| 2.82   |
| 2.76   |
| 2.85   |
| 2.94   |
| 3.04   |
| 3.13   |
| 3.05   |
| 3.13   |
| 3.15   |
| 3.02   |
| 3.02   |
| 3.02   |
| 3.09   |
| 3.07   |
| 3.13   |
| 3.20   |
| 3.17   |
| 3.17   |
| 3.19   |
| 3.15   |
| 3.10   |
| 3      |
| 3      |
| 2.95   |
| 2.92   |
| 2.94   |
| 2.92   |
| 2.89   |
| 2.88   |
| 2.85   |
| 2.89   |
| 2.89   |

| hy_oas |
|--------|
| 2.91   |
| 2.88   |
| 2.91   |
| 2.91   |
| 2.98   |
| 2.99   |
| 2.95   |
| 2.90   |
| 2.88   |
| 2.83   |
| 2.84   |
| 2.86   |
| 2.87   |
| 2.84   |
| 2.81   |
| 2.83   |
| 2.81   |
| 2.79   |
| 2.79   |
| 2.76   |
| 2.74   |
| 2.74   |
| 2.75   |
| 2.76   |
| 2.71   |
| 2.65   |
| 2.65   |
| 2.73   |
| 2.69   |
| 2.64   |
| 2.68   |
| 2.69   |

| hy_oas |
|--------|
| 2.71   |
| 2.72   |
| 2.77   |
| 2.80   |
| 2.88   |
| 2.81   |
| 2.85   |
| 2.86   |
| 2.97   |
| 2.87   |
| 2.84   |
| 2.86   |
| 2.84   |
| 2.92   |
| 2.95   |
| 2.94   |
| 2.94   |
| 2.86   |
| 2.88   |
| 2.86   |
| 2.95   |
| 2.97   |
| 2.94   |
| 2.98   |
| 3.10   |
| 3.12   |
| 3.03   |
| 3.08   |
| 2.97   |
| 3      |
| 3.13   |
| 3.19   |

| hy_oas |
|--------|
| 3.06   |
| 3.09   |
| 3.17   |
| 3.28   |
| 3.27   |
| 3.22   |
| 3.20   |
| 3.27   |
| 3.24   |
| 3.19   |
| 3.19   |
| 3.17   |
| 3.21   |
| 3.42   |
| 3.46   |
| 3.28   |
| 3.16   |
| 3.17   |
| 3.13   |
| 3.05   |
| 3.12   |
| 2.94   |
| 2.90   |
| 2.94   |
| 2.95   |
| 2.84   |
| 2.85   |
| 2.86   |
| 2.83   |
| 2.87   |
| 2.85   |
| 2.84   |

| hy_oas |      |
|--------|------|
|        | 2.86 |
|        | 2.86 |
|        | 2.84 |
|        | 2.85 |
|        | 2.82 |
|        | 2.83 |
|        | 2.77 |
|        | 2.78 |
|        | 2.77 |
|        | 2.75 |
|        | 2.79 |
|        | 2.81 |
|        | 2.79 |
|        | 2.82 |
|        | 2.82 |
|        | 2.76 |
|        | 2.80 |
|        | 2.83 |
|        | 2.86 |
|        | 2.80 |
|        | 2.78 |
|        | 2.74 |
|        | 2.74 |
|        | 2.72 |
|        | 2.71 |
|        | 2.72 |
|        | 2.72 |
|        | 2.74 |
|        | 2.72 |
|        | 2.71 |
|        | 2.75 |
|        | 2.74 |

| hy_oas |
|--------|
| 2.76   |
| 2.75   |
| 2.78   |
| 2.80   |
| 2.78   |
| 2.71   |
| 2.66   |
| 2.71   |
| 2.63   |
| 2.66   |
| 2.66   |
| 2.65   |
| 2.71   |
| 2.76   |
| 2.78   |
| 2.83   |
| 2.80   |
| 2.75   |
| 2.74   |
| 2.75   |
| 2.74   |
| 2.72   |
| 2.67   |
| 2.70   |
| 2.70   |
| 2.69   |
| 2.69   |
| 2.72   |
| 2.71   |
| 2.71   |
| 2.73   |
| 2.69   |

|  |  |  |  | hy_oas |
|--|--|--|--|--------|
|  |  |  |  | 2.69   |
|  |  |  |  | 2.68   |
|  |  |  |  | 2.77   |
|  |  |  |  | 2.79   |
|  |  |  |  | 2.81   |
|  |  |  |  | 2.84   |
|  |  |  |  | 2.87   |
|  |  |  |  | 2.84   |

Table 1. Credit spread series

| regime | count | avg_fwd_ret_... | median_fwd_ret_... | hit_rate_pct |
|--------|-------|-----------------|--------------------|--------------|
| Q1     | 106   | -1.83%          | -1.43%             | + 46.23%     |
| Q2     | 143   | + 2.56%         | + 4.04%            | + 74.13%     |
| Q3     | 157   | + 5.62%         | + 5.40%            | + 95.54%     |
| Q4     | 157   | + 7.26%         | + 6.89%            | + 100.00%    |
| Q5     | 158   | + 9.42%         | + 10.48%           | + 93.04%     |

Table 2. Forward returns by spread regime

Limitations

**NOTE**

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

regime\_fwd\_returns

| regime | count | avg_fwd_ret_... | median_fwd_ret_... | hit_rate_pct |
|--------|-------|-----------------|--------------------|--------------|
| Q1     | 106   | -1.83%          | -1.43%             | + 46.23%     |
| Q2     | 143   | + 2.56%         | + 4.04%            | + 74.13%     |
| Q3     | 157   | + 5.62%         | + 5.40%            | + 95.54%     |
| Q4     | 157   | + 7.26%         | + 6.89%            | + 100.00%    |
| Q5     | 158   | + 9.42%         | + 10.48%           | + 93.04%     |

spread

| observation_d... | hy_oas |
|------------------|--------|
|------------------|--------|

| observation_d... | hy_oas |
|------------------|--------|
| 2025-08-18       | 2.88   |
| 2025-08-19       | 2.90   |
| 2025-08-20       | 2.94   |
| 2025-08-21       | 2.95   |
| 2025-08-22       | 2.88   |
| 2025-08-25       | 2.80   |
| 2025-08-26       | 2.78   |
| 2025-08-27       | 2.78   |
| 2025-08-28       | 2.75   |
| 2025-08-29       | 2.82   |
| 2025-08-31       | 2.84   |
| 2025-09-01       | 2.84   |
| 2025-09-02       | 2.92   |
| 2025-09-03       | 2.88   |
| 2025-09-04       | 2.84   |

## Statistics Summary

| Key         | Value                                                     |
|-------------|-----------------------------------------------------------|
| study       | credit_spread_leading_indicator                           |
| category    | macro                                                     |
| status      | success                                                   |
| parameters  | {'lookback_years': 20, 'horizon': 63, 'benchmark': 'SPY'} |
| datasets    | ['credit_spreads/credit_spreads_features']                |
| rows        | 8155                                                      |
| git_commit  | 978cf61eefd9b125a5087073e70ef1c9d91f687b                  |
| executed_at | 2026-08-07T08:47:11.711140+00:00                          |

## Glossary

| Term      | Definition                                                     |
|-----------|----------------------------------------------------------------|
| n / count | Number of observations or events in the sample.                |
| edge      | Average excess return versus the unconditional baseline.       |
| hit-rate  | Share of events with a positive forward return.                |
| p-value   | Probability of observing the result under the null of no edge. |

| Term             | Definition                                    |
|------------------|-----------------------------------------------|
| *** / ** / *     | Significance at 99.9% / 99% / 95% confidence. |
| $\alpha$ (alpha) | Intercept or active return versus benchmark.  |
| $\beta$ (beta)   | Sensitivity to a reference factor or market.  |
| $\sigma$ (sigma) | Volatility / standard deviation of returns.   |

## Document Control

| Field                      | Value                                  |
|----------------------------|----------------------------------------|
| Report schema              | 2                                      |
| Framework version          | 0.1.0                                  |
| Classification             | internal                               |
| Generated                  | 2026-08-07 08:47 UTC                   |
| Author                     | Cerberus Research                      |
| Study slug                 | credit_spread_leading_indicator        |
| Run ID                     | 20260807_094711                        |
| Git commit                 | 978cf61e                               |
| Datasets                   | credit_spreads/credit_spreads_features |
| Parameters hash            | c5fdac1b6f11                           |
| Parameter · lookback_years | 20                                     |
| Parameter · horizon        | 63                                     |
| Parameter · benchmark      | SPY                                    |
| Data fingerprint           | d0e63220820a                           |
| Duration                   | 2.1s                                   |